MODEL QUESTION PAPER
B.Sc(Hons) Physics Major
MG5DSEPHY305 INTRODUCTION TO GROUP THEORY

Duration: 2 Hours **Max. marks : 70**

Part A

(Answer any TEN questions. THREE marks each)

1. Define group. Check whether the set of all integers form a group with addition as the rule of composition. [CO1]
2. State Cayley's theorem. [CO1]
3. Define reflection and inversion operations of a point group. [CO2]
4. Write down the group table of the point group D_2. [CO2]
5. Define a group representation. Write an example. [CO3]
6. Define character of a representation. [CO3]
7. When is a group representation irreducible? [CO3]
8. State the criteria for irreducibility of a group representation. [CO3]
9. Do the set of all translations on a plane form a continuous group? Is it abelian? [CO4]
10. Write an example of a rotation matrix. Show that it is orthogonal. [CO4]
11. Write down $SU(2)$ Lie algebra. [CO4]
12. What are weights of a $SU(2)$ representation? [CO5]

Part B

(Answer any 8 questions. FIVE marks each)

13. Define proper subgroup. Find proper subgroup of C_4. [CO1]
14. Using Lagrange's theorem, show that any group with order that is a prime number must be abelian. [CO1]
15. Write down the permutation operations of three objects and obtain the group table of permutation group S_3. [CO1]
16. Explain equivalent axes by choosing any point group as example. [CO2]
17. Explain how Uniaxial groups and Dihedral groups form groups with pure rotations. Give an example each. [CO2]
18. Give an example of point group containing rotation reflections. Write down its group table. [CO2]

19. State Schur's lemmas and explain its relevance for determining irreducibility. [CO3]

20. What is group algebra? How are the number of non equivalent irreducible representations of a group determined? [CO3]

21. Show that any rotation matrix can be expressed as $e^{i\mathbf{n}\cdot\mathbf{L}}$. You may choose $\mathbf{n}$ to be any convenient unit vector. [CO4]

22. Show that the $SU(2)$ generators $L_\pm = L_x \pm iL_y$ raises or lowers eigenvalues of L_z.[CO4]

QP Code **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) THEORY EXAMINATION OCTOBER 2026

V SEMESTER

DSE - MG5DSEPHY307- OP AMP AND LINEAR INTEGRATED CIRCUITS

(2024 ADMISSION ONWARDS)

Duration: 1 hour 30 minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Answer Questions

Answer any **10** questions. Each question carries **2** marks

1. State any four characteristics of ideal op-amp [K] / [CO1]
2. An op-amp has a differential gain 1500 and common mode gain 0.0075. Find the CMRR in decibels [U] / [CO1]
3. Define the concept of virtual ground in an inverting amplifier [U] / [CO1]
4. What is an instrumentation amplifier? Mention any two of its important features [U] / [CO2]
5. State the output relationship of an op-amp integrator for a sinusoidal input [A] / [CO2]
6. What is a current-to-voltage (I–V) converter? [U] / [CO2]
7. Write the output-voltage equation of an op-amp logarithmic amplifier [U] / [CO2]
8. What are astable and monostable multivibrators? [U] / [CO2]
9. Differentiate between voltage-series and voltage-shunt feedback [U] / [CO3]
10. Write the expression for input resistance of an amplifier with voltage-series feedback [U] / [CO3]
11. How does the voltage shunt feedback affect the input resistance of an amplifier [An] / [CO4]
12. An amplifier has an output resistance of 10 kΩ and loop gain Aβ=9. Calculate its output resistance with voltage shunt feedback [A] / [CO4]

[2x10 = 20]

Part B

Short Essay Questions

Answer any **5** questions. Each question carries **6** marks

13. With a neat circuit diagram derive the expression for the closed loop voltage gain of an inverting amplifier using the concept of virtual ground [U] / [CO1]
14. Explain the summing amplifier working in inverting configuration and obtain the expression for output voltage [U] / [CO1]
15 Explain the working of a three-op-amp instrumentation amplifier with a neat circuit diagram and state its applications. Obtain the expression for its output voltage [U] / [CO2]
16. An op-amp V–I converter is designed with an input voltage of 2 V and a resistance of 5 kΩ. a) Calculate the output current. b) If the input voltage is increased to 5 V, calculate the new output current [A] / [CO2]
17. With a neat circuit explain the operation of a triangular-wave generator [U] / [CO2]
18. Explain the operation of an op-amp differentiator and obtain the equation for output [U] / [CO2]
19. Derive the expression for the output resistance of a voltage series negative feedback amplifier [U] / [CO4]
20. Derive the expression for closed loop voltage gain of a voltage shunt feedback amplifier [U] / [CO3]

[6x5 = 30]